

FIFA World Cup 2026 Winner Prediction Using Machine Learning

Abstract

Football match prediction is a challenging problem due to the unpredictable nature of the sport and the influence of multiple performance factors. This project aims to analyze historical international football matches and build a machine learning model capable of predicting match outcomes. Using team rankings, Elo ratings, recent form, win percentage, and defensive statistics, a Random Forest Classifier was trained to classify match results as Home Win, Away Win, or Draw. The ultimate objective of this project is to provide a foundation for FIFA World Cup 2026 match and tournament predictions.

1. Introduction

The FIFA World Cup is the most prestigious football tournament in the world. Predicting match outcomes requires analyzing numerous factors including team quality, historical performance, recent form, and competitive strength.

The objective of this project is to:

- Analyze historical football match data.
 - Identify the most influential factors affecting match outcomes.
 - Build a machine learning model to predict match results.
 - Evaluate model performance using standard classification metrics.
 - Explore the feasibility of predicting FIFA World Cup 2026 outcomes.
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2. Dataset Description

The dataset contains 1,000 international football matches.

Each record includes information about:

- Home Team
- Away Team
- FIFA Ranking
- Elo Rating
- Recent Team Form
- Win Percentage

- Clean Sheet Percentage
- Host Advantage
- Continental Advantage
- Match Result

Target Variable

The target variable is Result:

Value Meaning

- 1 Home Team Win
- 0 Draw
- 1 Away Team Win

Dataset Distribution:

- Home Wins: 424
 - Away Wins: 414
 - Draws: 162
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3. Data Preprocessing

Several preprocessing steps were performed before model training:

Data Cleaning

- Missing values were identified and handled.
- Unnecessary columns were removed.
- Data types were verified and corrected.

Feature Engineering

Additional features were created to improve predictive power:

- Rank Difference
- Total Goals
- Team Strength Metrics

The final model utilized the following features:

- Home_Rank
 - Away_Rank
 - Home_Elo
 - Away_Elo
 - Home_Form
 - Away_Form
 - Home_Win_Pct
 - Away_Win_Pct
 - Clean_Sheet_Pct
 - Host_Advantage
 - Continent_Advantage
 - Rank Difference
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4. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to identify trends and relationships within the dataset.

The following analyses were performed:

- Match outcome distribution
- Team win percentage analysis
- Goal scoring analysis
- FIFA ranking comparison
- Elo rating comparison
- Correlation analysis
- Team performance analysis

Key Findings

- Teams with higher Elo ratings generally achieved better match outcomes.
- Recent form showed a strong relationship with match success.
- Historical win percentage contributed positively to performance.

- FIFA rankings alone were insufficient to explain all match outcomes.
 - Draws were relatively difficult to predict due to football's inherent uncertainty.
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5. Machine Learning Model

Model Selection

Random Forest Classifier was selected because:

- It handles non-linear relationships effectively.
- It is robust to noisy data.
- It provides feature importance analysis.
- It performs well on structured tabular datasets.

Train-Test Split

The dataset was divided into:

- Training Data: 80% (800 matches)
- Testing Data: 20% (200 matches)

This ensured the model was evaluated on unseen data.

6. Model Performance

Accuracy Score

The Random Forest model achieved:

Accuracy = 47%

While the accuracy is moderate, football prediction remains one of the most difficult predictive tasks due to the unpredictable nature of sporting events.

Confusion Matrix

The model performed reasonably well in identifying home and away wins but struggled to correctly predict draw outcomes.

This behavior is common in football prediction models because draws often occur due to unpredictable in-game events such as penalties, red cards, tactical changes, or individual moments of brilliance.

7. Feature Importance Analysis

The feature importance analysis revealed which variables contributed most to the model's predictions.

Feature	Importance
Away_Elo	0.112
Away_Form	0.111
Home_Elo	0.107
Home_Form	0.107
Home_Win_Pct	0.096
Clean_Sheet_Pct	0.092
Away_Win_Pct	0.090
Rank Difference	0.085
Away_Rank	0.081
Home_Rank	0.080
Continent_Advantage	0.019
Host_Advantage	0.019

Key Insights

1. Elo rating emerged as the strongest predictor of match outcomes.
2. Recent team form was nearly as important as Elo rating.
3. Win percentage significantly influenced predictions.
4. FIFA ranking contributed moderately to model performance.
5. Host advantage and continental advantage had minimal influence in this dataset.

8. Team Strength Analysis

A team strength table was created using aggregated historical statistics.

The analysis combined:

- Average FIFA Rank

- Average Elo Rating
- Average Recent Form
- Average Win Percentage

This provided a consolidated measure of team quality and allowed comparisons between national teams.

Example Findings

- Belgium recorded one of the highest win percentages.
- Austria achieved a strong average Elo rating.
- Brazil demonstrated strong recent form.
- Argentina maintained balanced performance across all metrics.

These observations support the importance of Elo ratings and recent form in football prediction.

9. Limitations

Several important real-world factors were unavailable in the dataset:

- Injuries and suspensions
- Player market value
- Squad selection
- Manager quality
- Tournament pressure
- Head-to-head records
- Home crowd influence

Additionally, the dataset size of 1,000 matches limits the model's ability to capture all football scenarios.

10. Future Improvements

Future enhancements may include:

- Expanding the dataset size
- Incorporating player-level statistics

- Adding Elo Difference and Form Difference features
- Implementing XGBoost and Gradient Boosting models
- Building a complete FIFA World Cup 2026 tournament simulator
- Generating match outcome probabilities

These improvements are expected to increase predictive accuracy and improve tournament forecasting capabilities.

11. Conclusion

This project successfully implemented a complete machine learning workflow for international football match prediction, including data collection, cleaning, exploratory data analysis, feature engineering, model development, and evaluation.

The Random Forest model achieved an accuracy of 47% on unseen match data. While football remains a highly unpredictable sport, the model successfully identified several important factors that influence match outcomes. Feature importance analysis revealed that Elo Rating, Recent Form, and Win Percentage were the most influential predictors of match results.

A team strength analysis was also conducted using historical performance metrics such as FIFA Ranking, Elo Rating, Recent Form, and Win Percentage. Among the analyzed teams, Argentina consistently demonstrated one of the strongest overall performance profiles, combining a high Elo rating, strong recent form, and a high historical win percentage.

Team	Avg Rank	Avg Elo	Avg Form	Avg Win %
Argentina	12.94	1971.58	3.54	63.42
Australia	11.52	1953.41	3.69	62.38
Austria	15.94	2000.23	3.42	61.65
Belgium	15.47	1979.30	3.20	64.37
Brazil	14.65	1970.31	3.67	60.23

Based on the combined analysis of team strength metrics and machine learning findings, Argentina emerges as the strongest candidate in this study for the FIFA World Cup 2026. The team's balanced performance across ranking, Elo rating, form, and winning consistency makes it the most complete contender among the analyzed nations.

Therefore, this project predicts Argentina as the most likely winner of the FIFA World Cup 2026, while acknowledging that real-world tournament outcomes may be influenced by injuries, squad selection, tactical decisions, and other unpredictable factors that are not captured within the current dataset.

Future work will focus on incorporating player-level statistics, advanced machine learning algorithms, and full tournament simulations to improve prediction accuracy and reliability.